Apertus for Cross-lingual Recognition of Semantic Differences (RSD

- **Track:** `Track 2A — UZH`
- **Event:** Online
- **Team:** `Gian Macagni, Tim Kessler`

Summary

In our project we show how Apertus 1.5 8B can be adapted to a fine-grained, low resource task, namely token-level recognition of semantic differences between related documents in a cross lingual setting. Our work builds on the benchmark defined by the SwissGov-RSD. Because there exist no large human-labelled training data, we decided to generate synthetic training data. We used the aligned SwissAdmin press-release corpus for its similarity of domain and because the required language pairs were already present in a similar style of translation. The words were labeled by an encoder model that we previously trained for the task on the projected SemEval-2016 iSTS corpus. For this we used KaLM-12B encoder and MLP heads as well as DiffAlign. With the resulting datasets we fine-tuned Apertus 1.5 8B with QLoRA and in a full train run with weighted loss.

The fine-tuned models were then evaluated on the SwissGov-RSD corpus via Spearman rank correlation on the gold labels. As a baseline, we also ran the evaluation on the base model (Apertus 1.5 8B). To refine the output and avoid invalid JSON responses from the model, we constrained the model output with llguidance grammar. As a basis for our evaluation prompt, we used the monolingual evaluation prompts from the SwissGOV-RSD paper and refined them into one multilingual prompt.

Jupyter notebooks were used for the full pipeline. Three make up the workflow for the creation of the synthetic data, two notebooks were used for training and another one for model evaluation. Besides that we also created a modified version of the original SwissGov-RSD evaluation prompt template and used enforced decoding generation with a special grammar to ensure valid JSON output.

